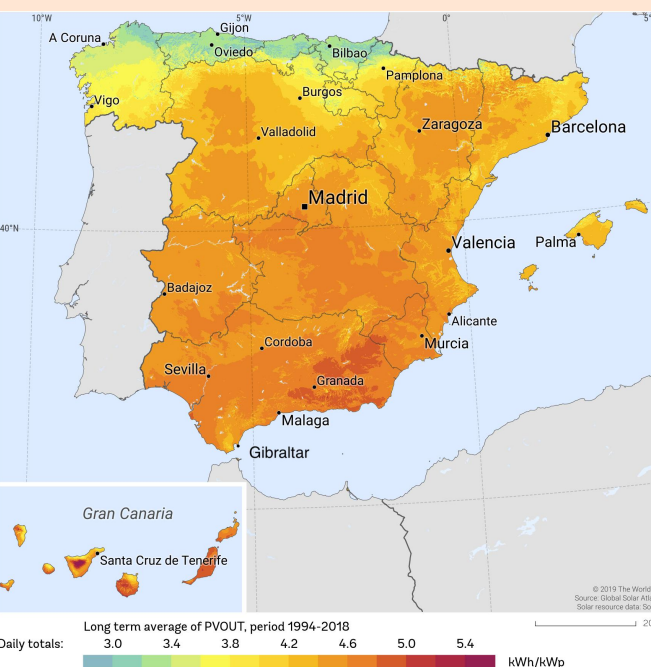
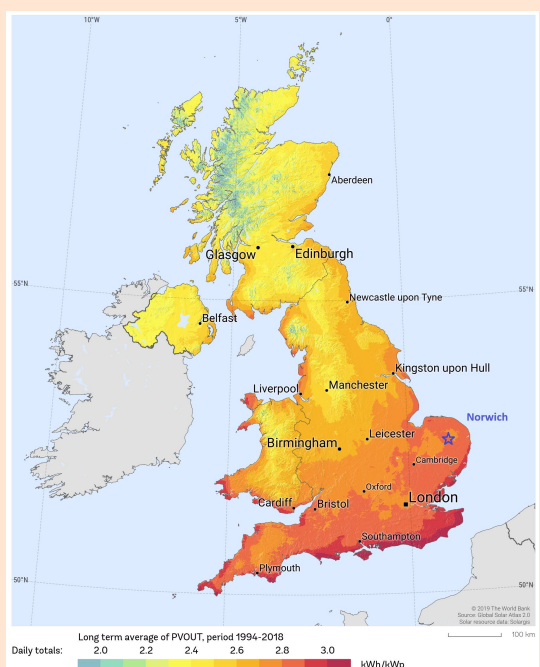


SUMMARY

- The average national PV potential (GHI) and solar thermal potential (DNI) of Spain are 68.1% and 120.2% respectively higher than the UK's. [1]
- UK falls behind Spain in terms of both market size, polit-economy, and future positivity, with a solar energy capacity difference of 35.6GW and 14.7GW for Spain and UK, respectively. [2][3]
- Average locations (Madrid & Barcelona) in ESP still outperform prime locations in the UK.
- Limited available locations for Gibraltar IAPV, yet shadowing presents in all available locations.
- High-performance result for Norwich Rooftop PV partially thanks to a small system size.
- Solar thermal is mostly used for CSP in Spain, whereas in the UK are mostly solar heaters.

SOLAR PV RESOURCES



a. UK Mainland [8]

	UK
PV Potential, City Avg. [8]	1643.5 kWh
TMY GHI, Site Avg. [1]	196.0 Wm ⁻²

b. Spain & Gibraltar [8]

	Gibraltar	Norwich	Madrid	Barcelona
PV Potential, City Avg. [8]	1643.5 kWh	1022 kWh	1643.5 kWh	1507.5 kWh
TMY GHI, Site Avg. [1]	196.0 Wm ⁻²	121.4 Wm ⁻²	207.2 Wm ⁻²	192.7 Wm ⁻²

PV MARKET & ECONOMIC POLICIES

UK, PV Market

Little potential

- 2nd most popular renewable energy market in the UK by annual/ total installation 2013-23. [3]
- 1GW capacity added 2022-23, i.e., only 2.5% of wind energy, and reached 15.7GW by 2023. [3]

Spain, PV Market

Greenlight ahead

- 1st/2nd largest market by annual/total installation capacity 2018-2023 in Europe. [9]
- 3.217GW capacity added in 2022 (40% of EU+UK's sum) and had 35.6GW in 2023. [10]
- 32.2GW before 2028, w/ per MWh wholesale skyrocketing from €51 to €180 in 2020-22. [9]

UK, Econ. Policies

Slow but steady

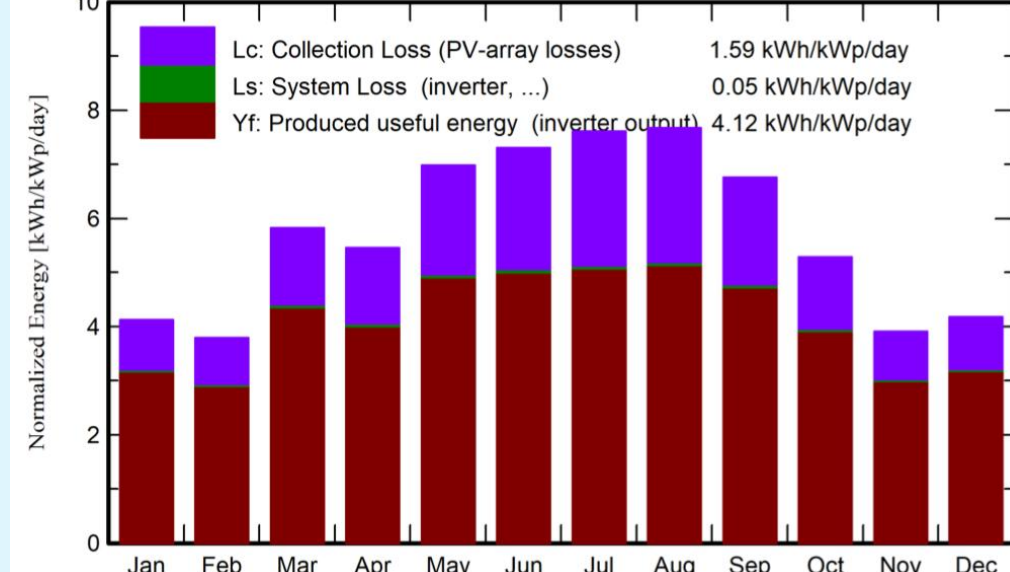
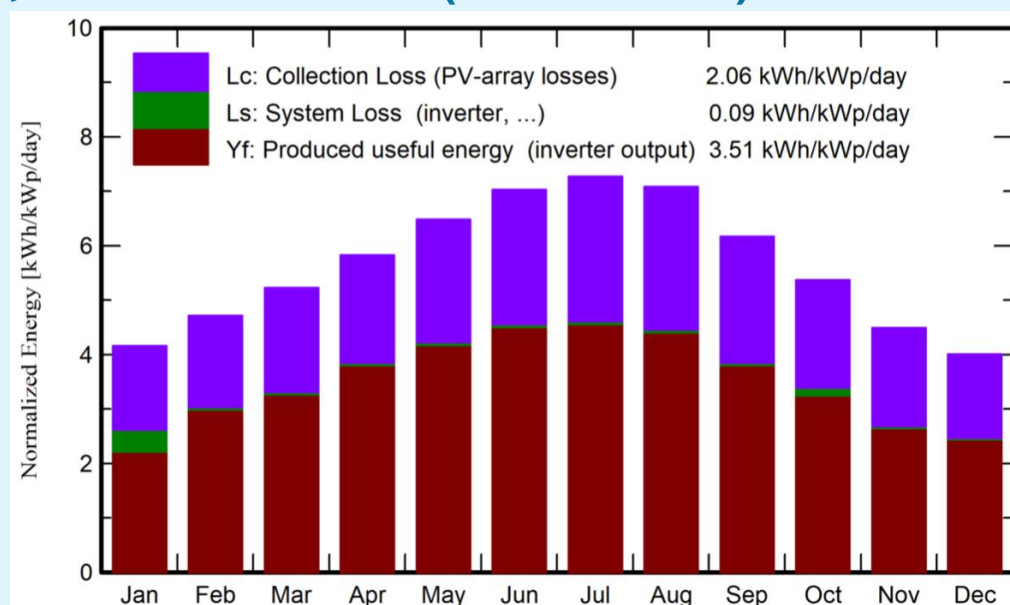
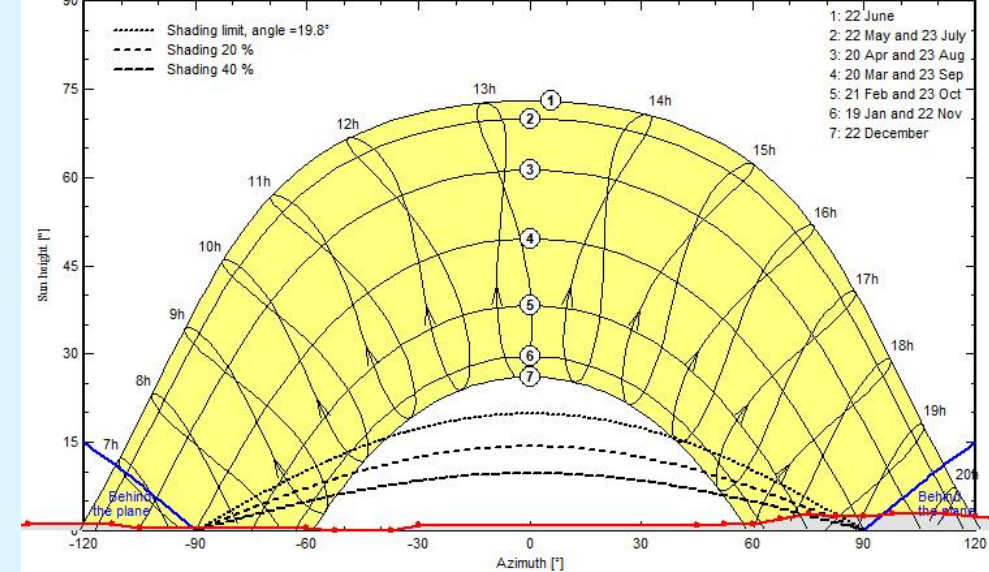
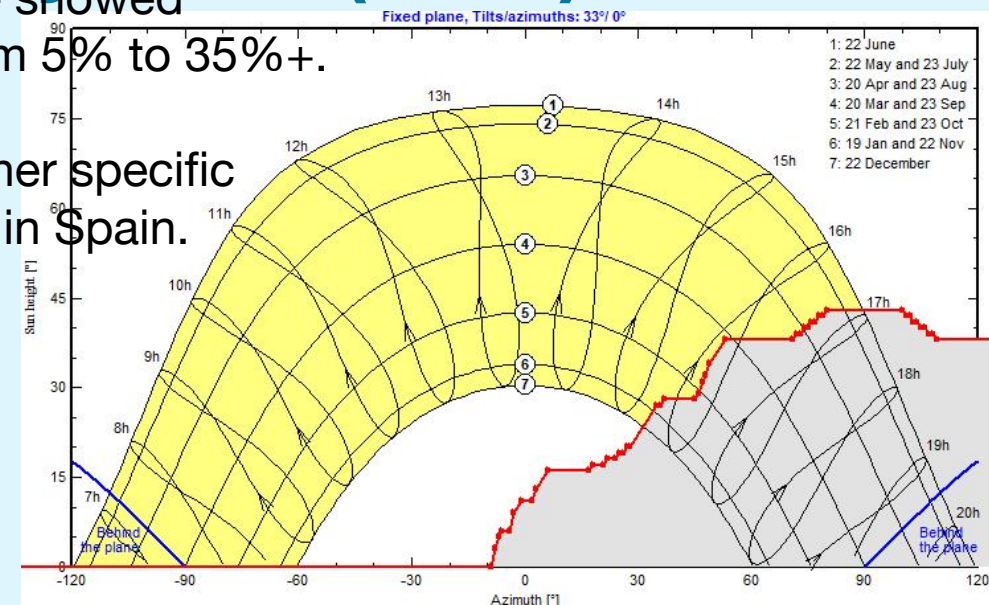
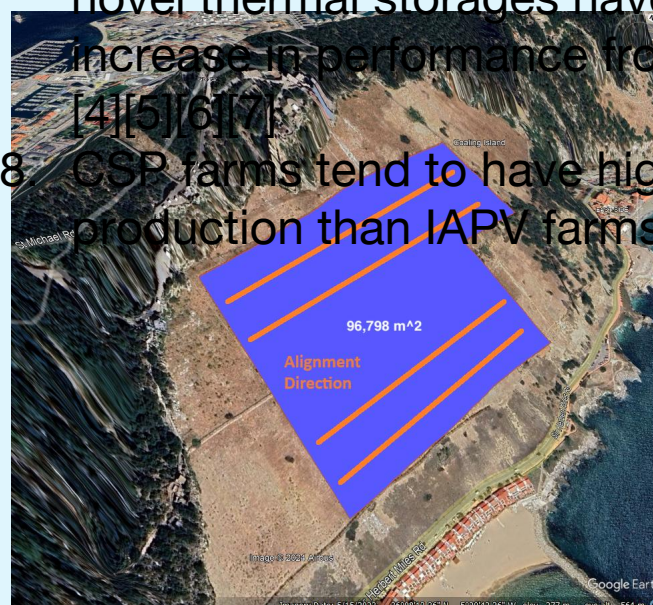
- INCENTIVE: SEG ~£0.15/kWh after tax, and negotiable in Gibraltar, UK. [11] [12]
- GRANTS & LOANS: Available £1bil from the UK Net-Zero-Innovation. [13]

Spain, Econ. Policies

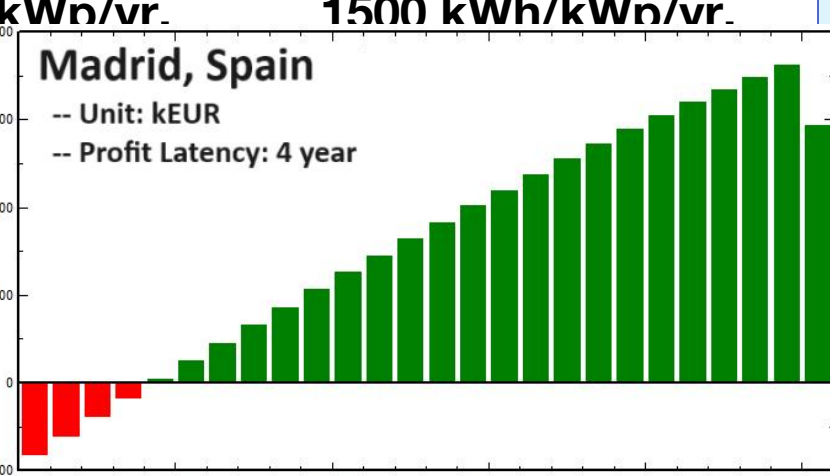
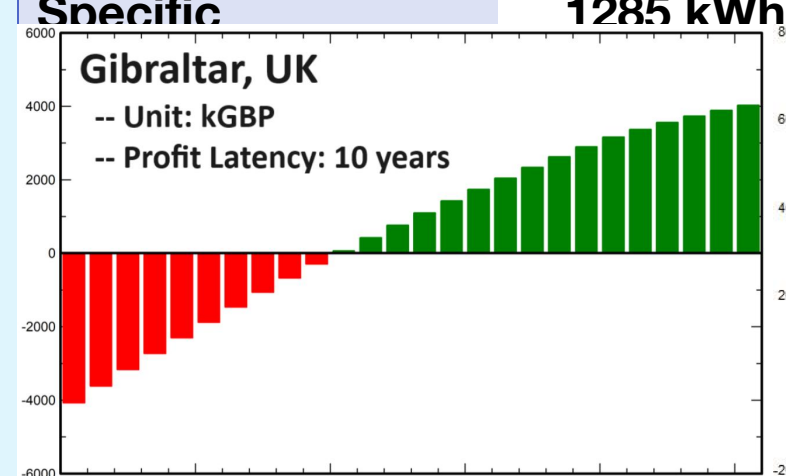
Cloudy to clear

- INCENTIVES: FIT €0.43/kWh, quota obligation ~€0.5/kWh, and tax exemptions. [10][14]
- GRANTS & LOANS: More than €165bil from RE-Power-EU and European Investment Bank. [15][16][17]

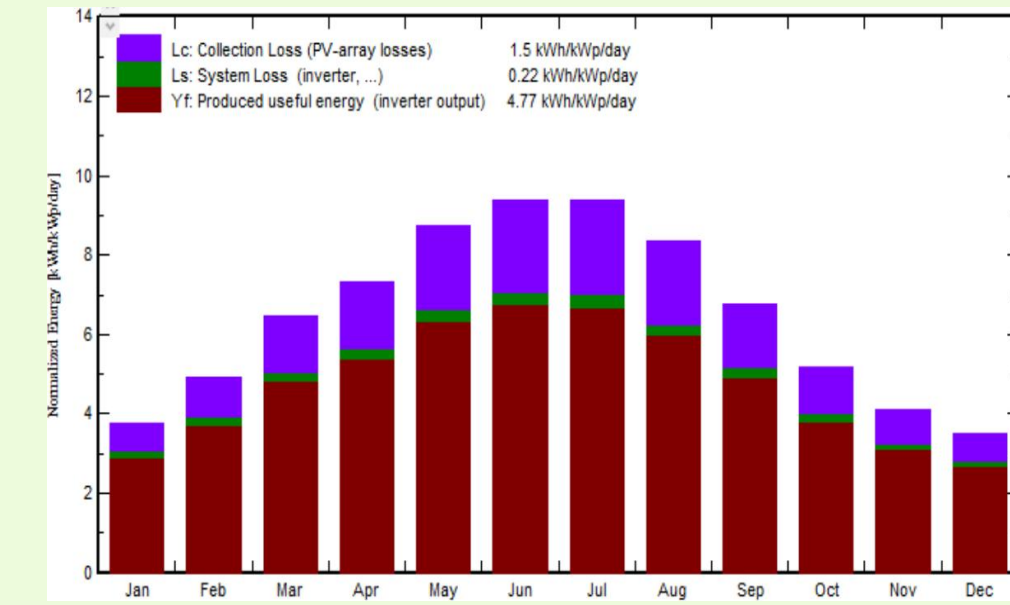
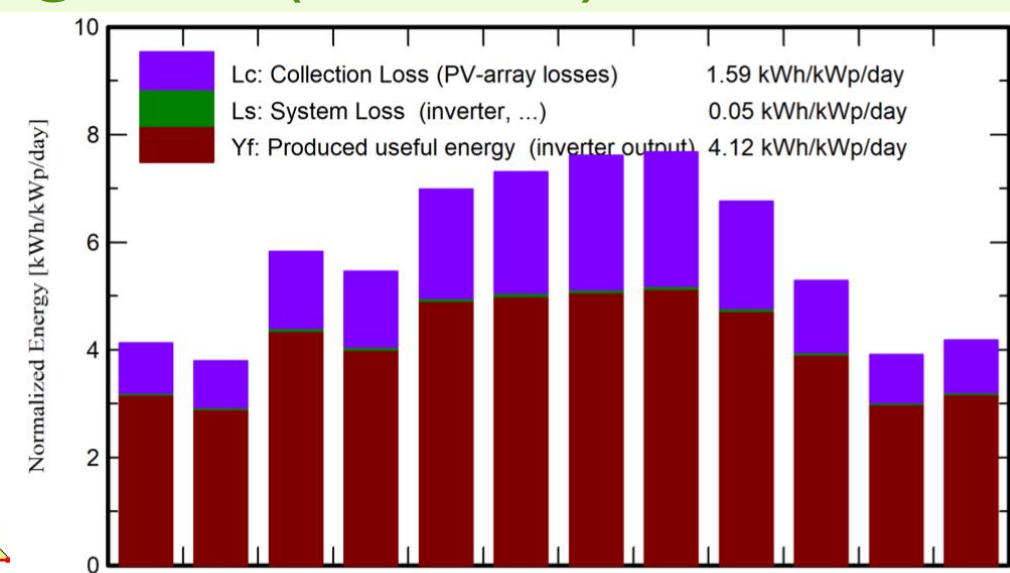
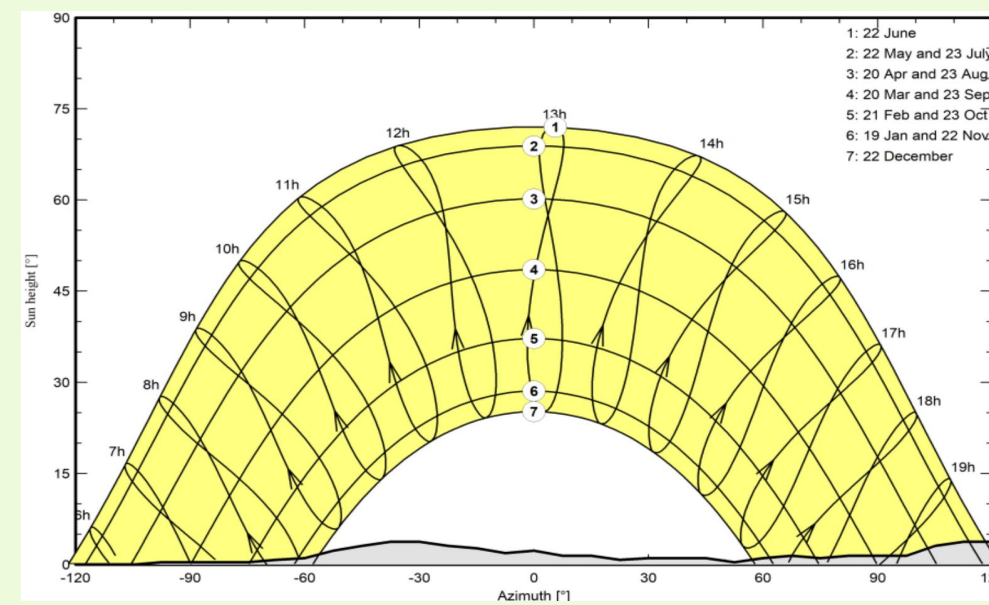
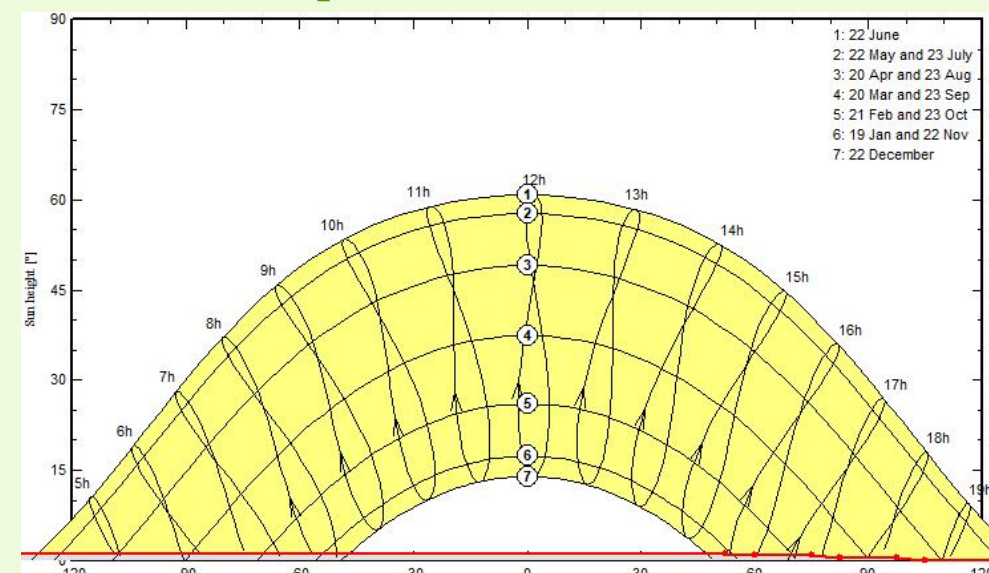
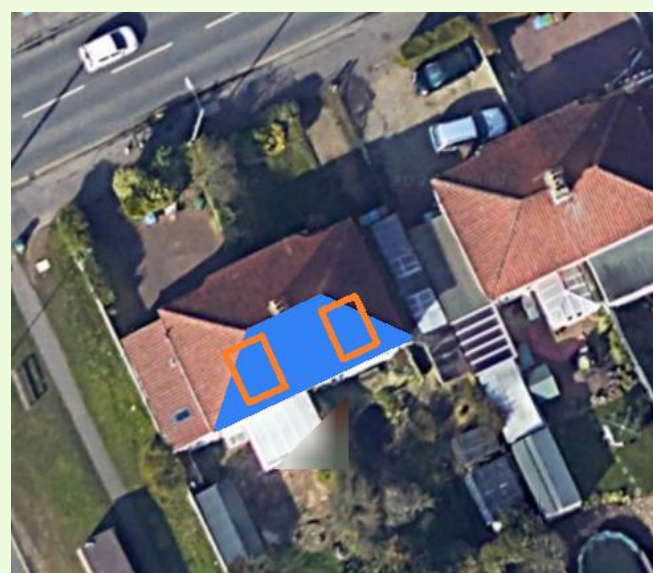
Rooftop PV (IAPV): La Caleta, Gibraltar (UPPER) VS Vicálvaro, La Ciudad de Madrid (LOWER)



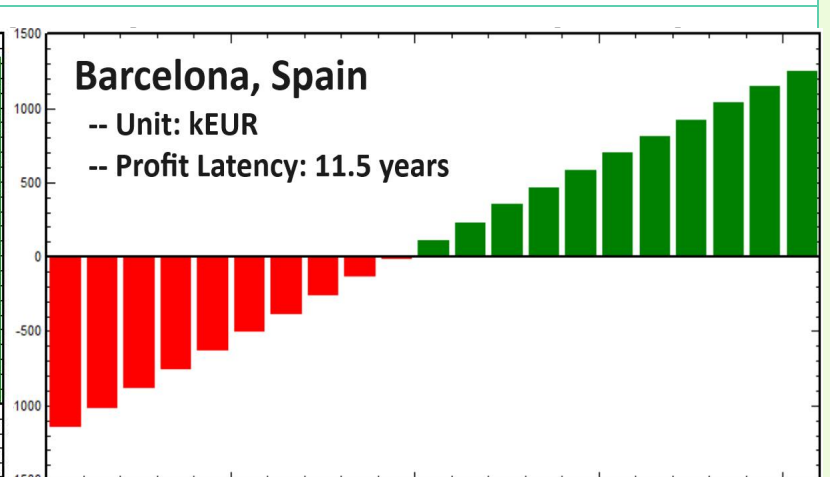
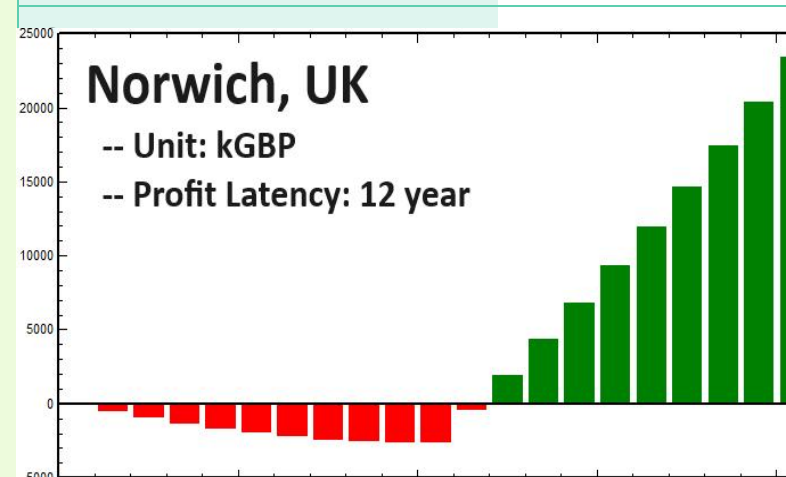
	Gibraltar (15.0 MW)	Madrid (132.7 MW)
Capacity	25,852*580 Wp Sharp NB-DJ580	228,781*580 Wp Sharp NB-DJ580
Inventor	4*3125 kW Sungrow 3125-HV-20	17*6874 kW Sungrow SG6250KVA
Tilting/Azimuth Angle	32.2/0 deg	28.0/0 deg
Actual/Panel Area	96,798/70,000 m2	2,711,086/700,000 m2
P75/P50 Production	18,800/19,271 MWh/yr.	196,900/199,077 MWh/yr.
Specific	1285 kWh/kWp/yr.	1500 kWh/kWp/yr.



Rooftop PV: Norwich, England (UPPER) VS Eixample, Barcelona (LOWER)

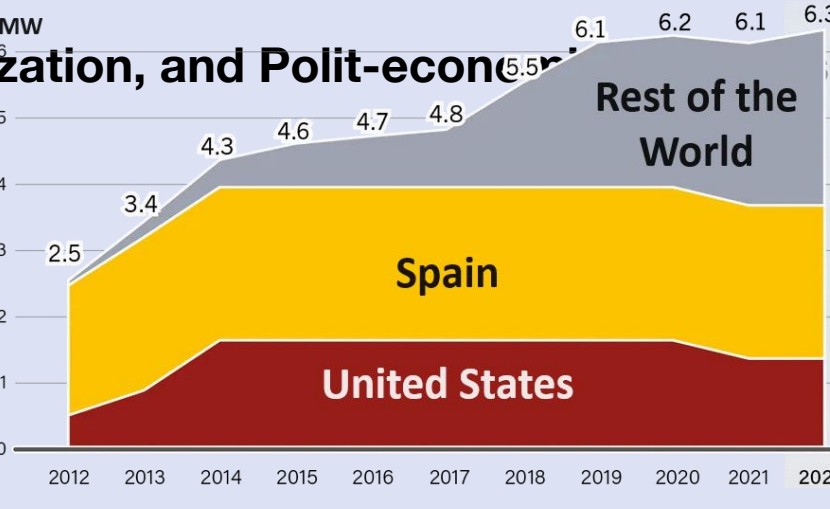
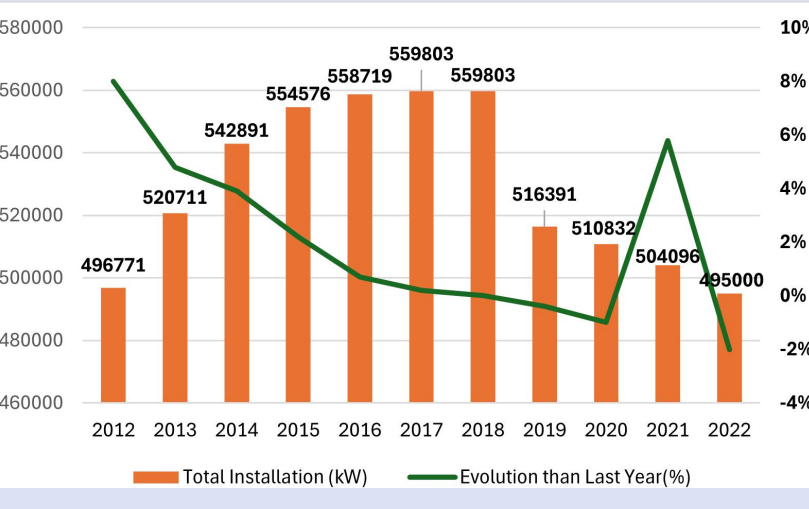


	Norwich (8.8 kW)	Barcelona (413 kW)
Capacity (kW)	16*0.55 Sharp NU-JD550	876*0.47 SunPower SPR-X21-470-COM
Inventor (kW)	2*4 SMA 4000-U28	2*2 ZigorSunzet 200KVAT
Tilting/Azimuth Angle	30/[-78, +22] deg	35/[-120, +120] deg
Panel Area (m2)	44.0	6962
Average Production	11,051 kWh/yr.	717 MWh/yr.



Solar Thermal Potential & Market: UK VS Spain Mainland

	UK Mainland	Spain Mainland
Major Solar Thermal Application, "STA"	Small/Mid-Scale Solar Thermal Heating (STH)	Concentrating Solar Power (CSP)
Average Solar Potential for that STA	75%-High GHI: 1004 kWh/m2 [8]	75%-High DNI: 2045 kWh/m2 [8]
Capacity of that STA	495 MW [18]	2,310 MW [2]
Comments	Smallest, in EU27+CH+UK	Largest, Worldwide
Future Market Growth	Decreasing Maintained	Slow Revival
Installation Incentives	Expired for Domestic	Feed-in-Premium-like



Notable CSP Projects in



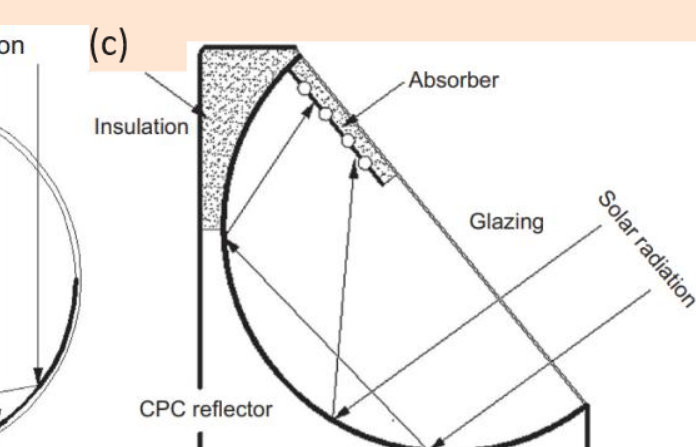
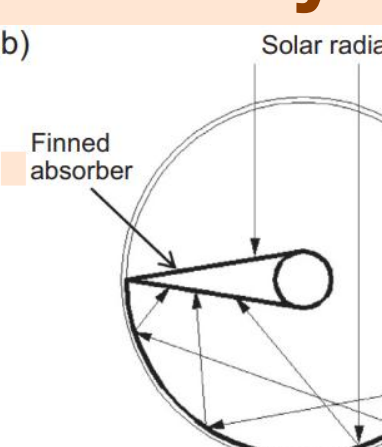
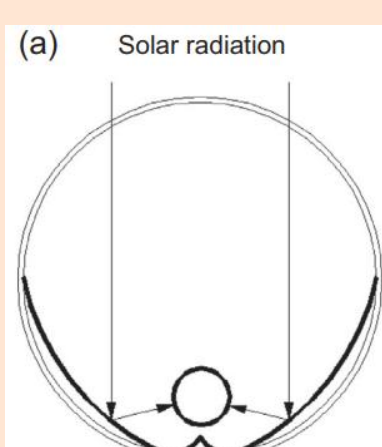
PS-10 Central Tower CSP, Andalucia: 11 MW· 75 000 m2· 624 heliostats· 21 M€

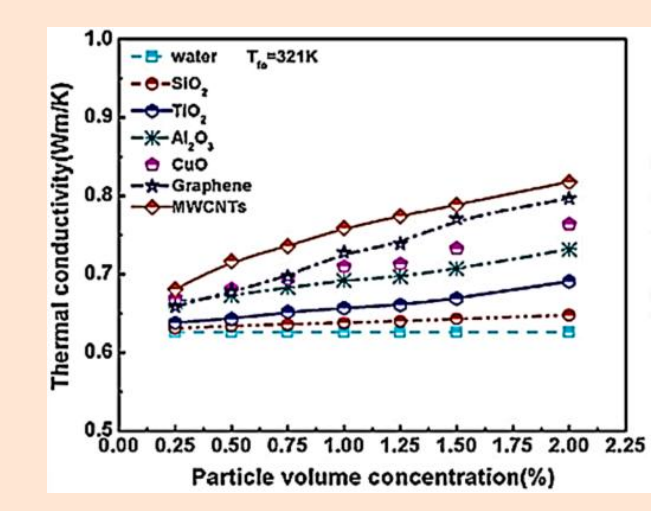


SYSTEM SITING REASONS

- Gibraltar is chosen for its best solar potential among all UK locations. The site sits on the unused, 33-deg slope next to La Caleta.
- Vicálvaro is an unutilized area between the Madrid city center and MAD airport, i.e. short cabling, and benefits from nearby highways.
- Eixample consists of blocks similar in heights and rooftop shapes, and every block are reasonably spaced between each other.
- Norwich is much less cloud covered than most UK municipalities.

Solar Thermal System: State of the





2) Nanofluids VS Water [5] 3) Solid State Energy Storage for CSP [6]